Ben Chapman-Kish

github.com/BenChapmanKish | ben-ck.com | linkedin.com/in/ben-chapman-kish

Skills

Languages: Python, Java, C++, C, Rust, JavaScript, Kotlin, Swift, Obj-C, VHDL, x86 Assembly, SQL, Regex, Shell
Tools: Git, JIRA, Docker, Postman, Wireshark, Netmap, CUDA, Quartus, Hyperfine, Flamegraph, Xcode, Unity
Libraries: Google Cloud Platform, Google ADK, TensorFlow/Keras, SciKit-Learn, matplotlib, numpy, pandas, sqlalchemy, Flask, unittest, asyncio, Node, Express, MySQL, PostgreSQL, MongoDB, Kafka, Spark, Thrift, Hadoop

Professional Experience

OwnersBox Fantasy Sports — Kitchener, ON
AI Team Lead — April 2025 – Present

- Orchestrated development of major AI project, taking it from the design stage to final product launch
- Spearheaded formation of AI team, defining team policies and procedures, identifying and proposing project opportunities, and hiring additional team members
- Managed AI project development by prioritizing and assigning tasks, collaborating with other team leads, managing timeline changes, and communicating status updates to company leadership
- Wrote comprehensive project proposal for major new initiative, complete with quantitative engineering analysis of design alternatives, well-defined constraints and criteria, stakeholder analysis, and risk analysis
- Oversaw development of ML orchestration workflows that enable a single developer to design, train, evaluate, and deploy new models at scale
- Created real-time data feed to manage constant changes in upcoming sports games, processing millions of updates per day from multiple input sources while filtering out noise and irrelevant data
- Designed complex cloud-based system that routes updated game data to thousands of individually hosted ML models that generate and store hundreds of thousands of predictions per day
- Optimized prediction system to minimize cloud compute costs and storage requirements while serving new predictions to production web servers within 3 seconds of receiving a game data update
- Developed internal employee chatbot using Google ADK, enabling non-technical teams to easily gather and analyze complex data queries and insights

University of Guelph — Guelph, ON
Teaching Assistant – Computer Organization & Design — January – April 2023

- Provided learning assistance to students both in-lab and via email, guiding students to debug code and determine submission requirements on their own as much as possible
- Designed term project requirements and created test cases for evaluating student submissions
- Fairly graded student lab and exam submissions by creating table of consistent responses and mistakes

FACEBOOK — (Remote) New York, NY
Software Engineering Intern — September – December 2020

- Improved ad matching throughput by up to 1,500x by re-designing indexing system to be locally-cacheable and require fewer database cross-references
- Configured profiling to identify bottlenecks in ad pipeline and monitor uptime across schema migrations

minted. San Francisco, CA

Software Engineering Co-op — January – April 2020

- Rewrote customizer system to reduce dependency on 3rd party services and improve load times by 30%
- Deployed advanced user segmentation, allowing company to market more relevant products to customers

Google Montreal, QC

Software Engineering Intern — May – August 2019

- Scaled debugging pipeline to handle 400x more traffic by migrating to better-suited data store and optimizing SQL queries, allowing engineers to quickly identify and track a greater variety of bugs
- Reduced maintenance costs by consolidating pipeline logic into smaller modules with full test coverage

toast Boston, MA

Software Engineering Co-op — September – December 2018

- Solved concurrency bug by designing thread-safe system for calculating prices, reducing crashes by 12%
- Created tool with Gradle and JDBC to replicate production data in a local environment, reducing time for developers to find and address customer-reported problems

TRIBAL SCALE Toronto, ON

Agile Software Engineer — January – April 2018

- Reduced dependency on server in radio app, decreasing load on server and improving load times by 55%

Defence Research and Development Canada Toronto, ON

Mobile Application Developer — May – August 2017

- Solved device portability issues with high profile mobile applications, enabling widespread deployment

UNIVERSITY of GUELPH Guelph, ON

Machine Learning Research Assistant — July – August 2016

- Trained deep learning models with Caffe to achieve optimal performance for facial pose estimation
- Designed and programmed framework for crowd-controlled gaming using Python and OpenCV

Leadership Experience

University of Waterloo ECE Student Society — 2019 – 2021

Founder, Co-President, Peer Mentor

- Founded and managed student society for undergrad ECE students to foster program cohesion, provide academic and professional supports, and facilitate knowledge-sharing across cohorts and years
- Planned team-building social events for students to encourage collaboration and communication
- Mentored lower-years academically with lab/project guidance, and professionally with resume critiques

University of Waterloo Engineering Society — 2016 – 2022

Class Representative, Orientation Week Leader & Director

- Acted as class representative in general meetings, advocating for solutions to challenges faced by ECE students and collaborating with representatives from other disciplines to achieve common goals
- Managed teams of orientation leaders during annual engineering orientation, co-ordinating team schedules during events, resolving conflicts between front-line leaders, and enforcing safety rules for first-years
- Created event plans and contingencies for components of orientation, supervising timing and operation

Personal Projects

QuickPic **2018**

- Created social network with iOS client that can take, edit, and send pictures to other users
- Implemented backend server with Node and Express to manage users and facilitate user interaction

Pebble Apps **2015 – 2016**

- Built smartwatch apps in C using event-driven programming to interact with sensors and actuators

Academic Projects

Identifying Mental Health Disorders on Twitter using LSTM Models **2024**

- Investigated the use of causal deep learning models to predict occurrences of mental health disorders based on a Twitter user's entire post history rather than individual tweets
- Employed RNN and LSTM models as they are causal neural networks good for memory-based NLP
- Used the Twitter-STMHD dataset, vectorizing tweet text with the Word Embeddings method

Classifying Music Genres using Hybrid CNN-RNN Models **2022**

- Explored the use of convolutional and recurrent models, either sequentially (CRNNs) or in parallel (PR-CNNs), to perform music genre recognition on input audio files
- Used the FMA dataset, vectorizing audio signals with Mel-frequency cepstral coefficients as network input

Training Intelligent Game Agents using Double Deep Q-Learning **2022**

- Applied deep reinforcement learning to complex games that require higher-level decision making and the ability to learn game mechanics through experience
- Modelled Super Mario Bros (1985) as an RL problem in order to train a game agent to play it, using Double Q-Learning with Deep Neural Networks for off-policy model-free learning

Matching Applicants to Jobs with a Distributed Network of Dynamically-Selected ML Servers **2022**

- Designed a distributed dynamic machine learning platform that learns from user reviews to match applicants with job openings on a professional network such as Glassdoor/LinkedIn
- Implemented custom data and transport layers to maximize service throughput and minimize latency
- Created custom fault-tolerant load balancer that can connect to new server instances while running, and selects the best backend server based on speed, uptime, and model accuracy for a given task
- Wrote custom ML servers that each have partial database caches kept in sync by the master server, that each contain several instances of neural networks that are optimized for different tasks

Education

University of Guelph **2022 – 2024 (on leave)**

Master's of Computer Engineering, Artificial Intelligence specialization

University of Waterloo **2016 – 2022**

Bachelor of Honours Computer Engineering

Awards

Recipient ----------- **University of Waterloo President's International Experience Award** **2018, 2020**

Winner ------------- **University of Waterloo ECE Design Days** **2017**

- Designed Arduino-powered ball launcher and wrote an Android app to control it

Top 10 Finalist --- **University of Waterloo EngHack hackathon** **2017**

- Created Android app for coordinating study groups featuring Facebook login and Firebase database